

2 AI Fundamentals Quiz ANS

1. Which definition of AI was adopted in the presentation?

- A) Computer systems performing tasks associated with intelligence, such as problem solving and decision making
- B) Any electronic spreadsheet
- C) Only humanoid robots
- D) A database with no predictive capability

ANS: A) Computer systems performing tasks associated with intelligence, such as problem solving and decision making

2. What is narrow AI?

- A) AI that performs virtually every human task
- B) AI designed to perform specific tasks
- C) AI with industrial-revolution-level impact by definition
- D) AI that outclasses humans at all intellectual tasks

ANS: B) AI designed to perform specific tasks

3. What distinguishes an API model from an open model?

- A) An API model cannot be used
- B) An API model publishes all weights
- C) An API exposes controlled queries without giving users the underlying model
- D) An API model has no restrictions

ANS: C) An API exposes controlled queries without giving users the underlying model

4. Which term describes an AI system with impact comparable to the Industrial Revolution?

- A) Narrow AI
- B) Closed AI
- C) Anomaly detector
- D) Transformative AI

ANS: D) Transformative AI

5. What is a modality in machine learning?

- A) A type of input data, such as text, images, or audio
- B) A model's legal license
- C) A hardware failure
- D) A fixed output threshold

ANS: A) A type of input data, such as text, images, or audio

6. Which task predicts categories from input features?

- A) Regression
- B) Classification
- C) Compression
- D) Clustering only

ANS: B) Classification

7. Which task predicts numerical outputs from input features?

- A) Classification
- B) Tokenization
- C) Regression
- D) Authentication

ANS: C) Regression

8. What does generative modeling do?

- A) Deletes all training examples
- B) Predicts only binary labels
- C) Prevents new samples
- D) Creates new data resembling the training data

ANS: D) Creates new data resembling the training data

9. Which sequence best describes a typical ML pipeline?

- A)** Collect data, label, choose architecture, train, evaluate, deploy
- B)** Deploy, then define the task
- C)** Evaluate before collecting data
- D)** Remove data and skip training

ANS: A) Collect data, label, choose architecture, train, evaluate, deploy

10. What does precision measure?

- A)** The fraction of all predictions that are correct
- B)** The purity of the examples flagged as positive
- C)** The proportion of relevant examples found
- D)** The time needed for inference

ANS: B) The purity of the examples flagged as positive

11. What does recall measure?

- A)** The model's memory capacity
- B)** The purity of flagged examples
- C)** The proportion of relevant examples successfully flagged
- D)** The number of training epochs

ANS: C) The proportion of relevant examples successfully flagged

12. Which objective concerns the speed at which a model produces results?

- A)** Transparency
- B)** Reliability
- C)** Scalability
- D)** Inference latency

ANS: D) Inference latency

13. What type of data does supervised learning use?

- A)** Labeled examples
- B)** Only unlabeled examples
- C)** Only agent-generated rewards
- D)** No data

ANS: A) Labeled examples

14. What is the main goal of unsupervised learning?

- A)** Follow explicit output labels
- B)** Discover patterns and relationships in unlabeled data
- C)** Maximize a reward through actions
- D)** Compile source code

ANS: B) Discover patterns and relationships in unlabeled data

15. How does reinforcement learning obtain learning signals?

- A)** Only from static labels
- B)** Only from clustering
- C)** From an agent's interaction with an environment and resulting rewards
- D)** From manual rules alone

ANS: C) From an agent's interaction with an environment and resulting rewards

16. What does self-attention encode?

- A)** Only the first element of a sequence
- B)** Hardware clock speed
- C)** A fixed label dictionary
- D)** Relationships and relative importance among sequence elements

ANS: D) Relationships and relative importance among sequence elements

17. What is the central architecture introduced for modern sequence modeling?

- A)** Transformer
- B)** Decision stump
- C)** Hash table
- D)** Relational database

ANS: A) Transformer

18. What do deep-learning scaling laws describe?

- A)** A guarantee of zero error
- B)** Predictable relationships among performance, compute, data, and model size
- C)** A ban on larger models
- D)** Only the price of hardware

ANS: B) Predictable relationships among performance, compute, data, and model size

19. In reinforcement learning, what is a policy?

- A)** A labeled training set
- B)** A model-access license
- C)** A mapping from states or observations to actions
- D)** A fixed reward value

ANS: C) A mapping from states or observations to actions

20. What defines the full reinforcement-learning problem?

- A)** One isolated prediction with no feedback
- B)** Only supervised labels
- C)** No environment
- D)** An agent repeatedly acts, receives observations and rewards, and influences future states

ANS: D) An agent repeatedly acts, receives observations and rewards, and influences future states